Digital Systems

Minor 3

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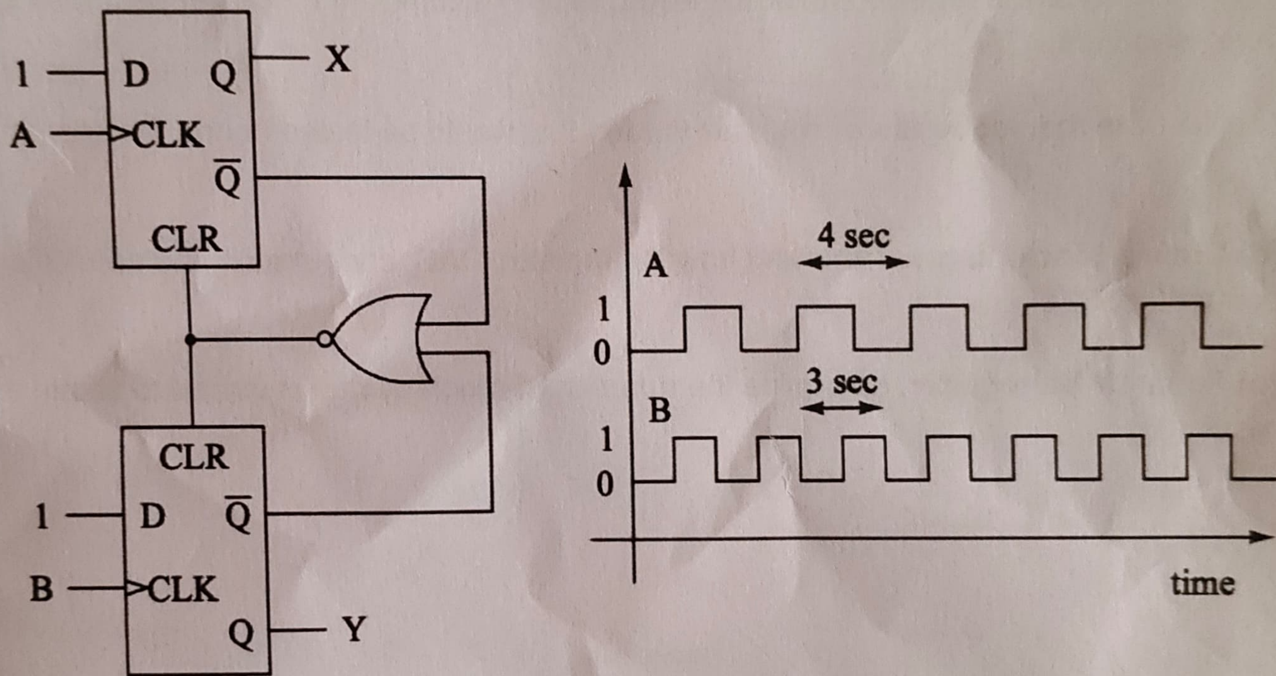
1. Construct a set-reset latch using only NAND and NOR gates. Write out its truth table. [2.5x4]
2. Design a sequential circuit to go through the sequence 0,5,2,1,3 using T Flip Flops. [15]
3. Consider the circuit diagram below. The signals A and B are as shown in the diagram. A has a period of 4 seconds, B has a period of 3 seconds. Assume both flip-flops to be cleared initially. Evaluate what X and Y are going to be, as functions of time. Can you draw something similar to a state diagram corresponding to the circuit? [5+5]



4. **Question: Digital Search Using a Comparator**

An unknown voltage **Vin** lies between **0 and Vref**. You are given the following components:

- A comparator
- A 4-bit register
- A 4-bit Digital-to-Analog Converter (DAC)

The DAC converts the value stored in the register into a voltage **Vdac**.

The comparator produces the following output:

- Output = **1** if **Vin ≥ Vdac**
- Output = **0** if **Vin < Vdac**

The goal is to determine the **4-bit digital value corresponding to Vin** using the **minimum number of comparison steps**.

Answer the following:

(a) Describe a **step-by-step procedure** to determine the correct 4-bit value stored in the register using the comparator output. [8]

(b) Which bit of the register should be tested **first** to minimize the number of steps? Explain your reasoning. [2]

(c) Illustrate the **sequence of register values** that would be tested during the search process. [4]

(d) Draw a **block diagram** showing how the register, DAC, comparator, and control logic are connected.[8]

(e) For an **N-bit register**, determine the **number of clock cycles required** to obtain the final result.[3]